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# The Pulmonary Agent-based Infection simulator (PAI): A Multi-Scale Agent-Based Model of Pulmonary Host-Pathogen Interactions

**Henrique AL Ribeiro** [1¶], **Sandra A Tsiorintsoa** [1], and **Reinhard Laubenbacher** [1]

**1** Department of Medicine, Division of Pulmonary, Critical Care, and Sleep Medicine, University of Florida, Gainesville 32610, FL, USA. ¶ Corresponding author

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## Summary

Understanding the spatial and regulatory dynamics of the immune response to pulmonary infections remains a central challenge in computational immunology. While mechanistic models, such as ODEs and PDEs, offer insights into intracellular processes, they are limited in representing discrete behaviors and spatial heterogeneity. The pulmonary Agent-based Infection simulator (PAI) addresses this gap through an extensible, multi-scale agent-based modeling designed to simulate lung immune responses to pathogens such as *Aspergillus fumigatus*.

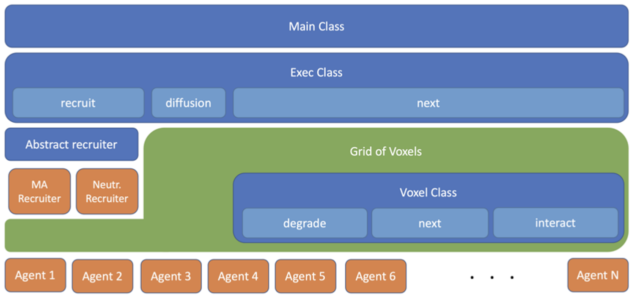
PAI operates within a three-dimensional voxelized space that approximates the alveolar microenvironment, incorporating host and fungal cells. It models inter- and intracellular signaling, immune cell recruitment and movement, and pathogen nutrient acquisition.

PAI is implemented in both Java (jPAI) and C++ (PAI++), with identical outputs and comparable performance, though PAI++ significantly reduces memory usage. With its modular architecture and biological fidelity, PAI provides a valuable tool for conducting in silico experiments where empirical approaches are limited.

## Statement of Need

Invasive Pulmonary aspergillosis is a human infection with increasing incidence in immunosuppressed patients such as those receiving chemotherapy or organ transplants (Pappas et al. (2010)). It has also been observed that 10%-14% of COVID-19 patients in the ICU develop invasive aspergillosis (Mitaka et al. (2021); Chong & Neu (2021)). Despite advances in diagnostics and therapy, mortality remains as high as 30–60% in recent surveys (Neofytos et al. (2013)).

Understanding the complex interplay between immune cells, pathogens, and signaling molecules during pulmonary infections requires computational models that capture both the spatial and regulatory dynamics of the immune response (Minucci et al. (2020); Yue & Dutta (2022)). While mechanistic models such as ODEs or PDEs are effective at simulating intracellular or population-level behavior, they lack the flexibility to represent discrete cellular interactions and heterogeneous spatial environments.

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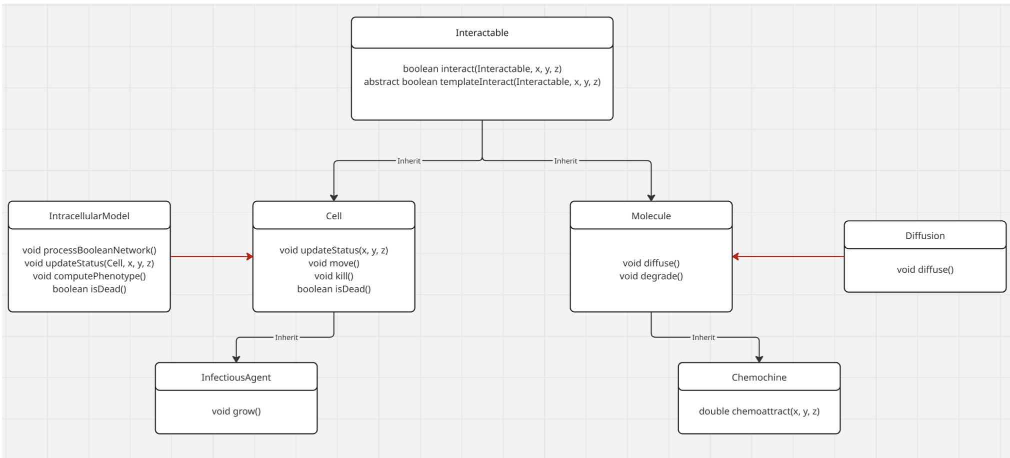


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**Figure 1:** Figure 1: ABM Architecture. A: Overall architecture. The "Exec" iterates through the grid-of-voxels (green) and executes the public methods of Voxel (blue). The Voxels iterates through their local lists of agents (orange bottom), executing their public methods. Exec also executes molecule diffusion, "garbage collection (gc)" of dead cells, and cell recruitment. B: Diagram showing the agent's classes. Black arrows represent inheritance, while red represents composition. All agents are interactables, which means that they can potentially interact with any other agents.

PAI addresses this gap by providing a scalable, extensible agent-based modeling framework for simulating the tissue-scale immune response to lung infections, such as *Aspergillus fumigatus*. PAI has been used before, and the mathematical and biological components are better explained in Ribeiro et al. (2022). While in Ribeiro et al. (2023), we modified it to simulate and generate a hypothesis about Covid-Associated-Pulmonary-Aspergillosis. Qu et al. (2025) expanded the original model to simulate the effect of NETosis and hemorrhage on Aspergillosis. However, none of these papers provides detailed information about the software.

## State of the Field

Several ABMs of the Immune System have been proposed. For example, the Celada-Seiden (Celada & Seiden (1992)) model and its successors, such as C-IMMSIM (Rapin et al. (2010)), introduced the concept of representing immune cells and molecules as discrete interacting agents governed by probabilistic rules. C-IMMSIM (Basak et al. (2021)) has been widely used to study vaccination strategies and systemic immune responses. However, its design is largely focused on systemic and lymphoid processes rather than the lung environment. In contrast, GRAN-SIM (Segovia-Juarez et al. (2004)) represents a highly specialized and biologically detailed ABM tailored to the study of tuberculosis granulomas. Its high level of specialization enables deep mechanistic insights but also limits its generalizability.

The PAI framework addresses this gap by integrating a spatially explicit, voxel-based representation of the lung with a modular and extensible agent architecture. Unlike general immune simulators, it explicitly models the pulmonary microenvironment, and unlike highly specialized models, it is structured to support adaptation to multiple pathogens and biological hypotheses.

# Research Impact Statement

Pulmonary infections such as invasive aspergillosis and viral pneumonias remain major causes of morbidity and mortality, particularly in immunocompromised populations (Pappas et al. (2010)), yet their underlying immune dynamics are difficult to study experimentally due to spatial complexity and limited access to human lung tissue. The Pulmonary Agent-based Infection simulator (PAI) addresses this challenge by providing a biologically grounded, spatially explicit computational framework that enables in silico experimentation on host–pathogen interactions within the lung microenvironment. PAI has been applied in hypothesis generation, experimental design, and therapeutic exploration, including the study of immune modulation, co-infections, and treatment strategies (Ribeiro et al. (2022); Ribeiro et al. (2023); Qu et al. (2025)).

# Software Design

## Architecture

As shown in Figure 1A, at the top level, a Main module orchestrates the initialization and execution of simulations, including setting up initial conditions, managing output, and controlling the simulation loop. Initialization is handled by a dedicated Initializer class, which instantiates the molecular fields, populates the simulation space with agents, and assigns intracellular models where appropriate.

The simulation domain is discretized into a three-dimensional voxel grid, where each voxel represents a microenvironment (~40 µm per side) and maintains localized lists of agents. The Voxel class governs local interactions (interact method), molecular degradation (degrade method), and cell updates (update method) and is invoked iteratively by the Exec class. The Exec class also handles molecular diffusion (via PDE solvers), cellular recruitment (based on chemokine gradients), and agent-level garbage collection.

Agents in the PAI are broadly categorized as Cells or Molecules, both inheriting from an abstract Interactable class (Figure 1B). This base class implements a symmetric interact method based on an "inductive" architecture: new agent types define how they interact with existing agents. If Agent "A" does not provide code to interact with Agent "B," nor does Agent "B" provide code to interact with "A," they learn on the fly that they do not interact (Figure 2).

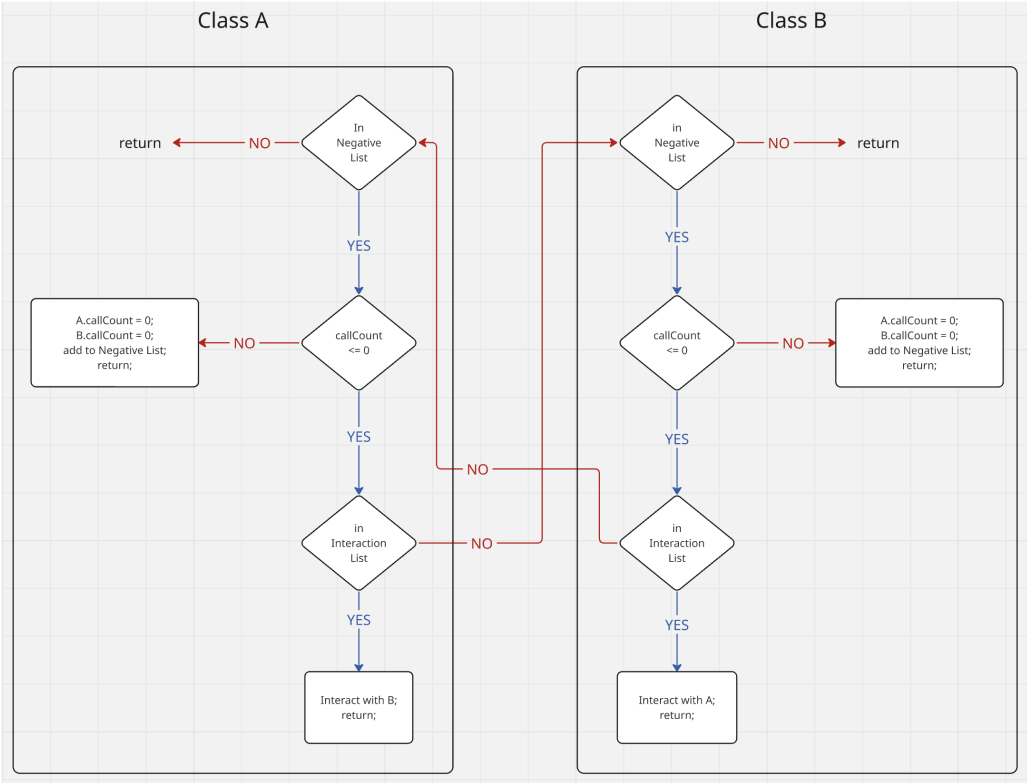


**Figure 2:** Figure 2: Dynamics of interaction between two agents. Decision diagram made by the methods "interact" and "templateInteract" from the two classes trying to interact ("A" and "B").

## Instantiation

A typical instantiation of the PAI model, as used in Ribeiro et al. (2022), involves a 10×10×10 voxel grid populated with both cellular and molecular agents. The simulation initializes molecular fields, Transferrin, Lactoferrin, TAFC, Iron, IL-6, TNF-α, IL-10, TGF-β, CXCL2, CCL4, and Hepcidin. It is also initialized with 640 Type II pneumocytes, 15 macrophages, 15 neutrophils, and 1920 *Aspergillus fumigatus* (in the resting conidia state) randomly distributed across the grid. The simulation proceeds via the Exec class, which iterates over all voxels and over the lists of agents they contain. Simulations typically run for 2160 iterations (approximately 72 hours), as in Ribeiro et al. (2022), where results quantitatively compared with *in vivo* data and demonstrated close agreement.

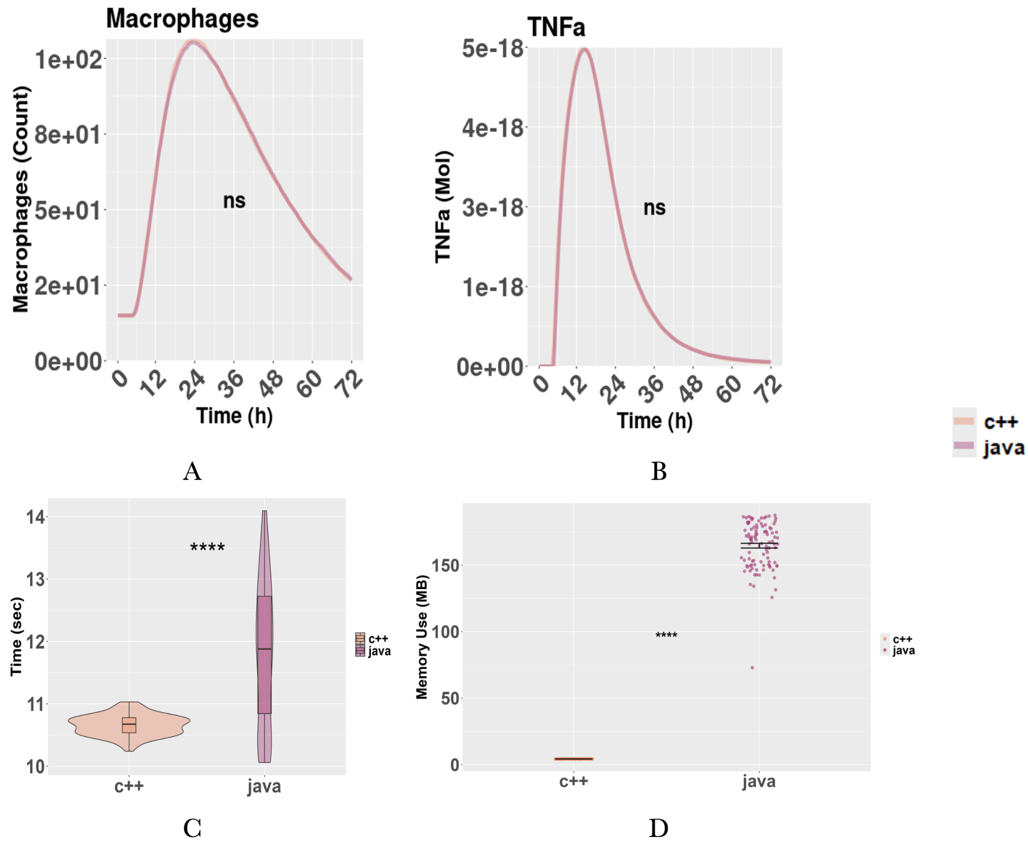


**Figure 3:** Figure 3: jPAI and PAI++ comparison. A: Macrophage time series. B: TNF-a time series. C: Running time. D: Memory usage. Memory recorded with Valgrind (C++) and JProfiler (Java). All figures represent the average of 100 simulations. Simulations initialized with 10X10X10 voxels, 640 pneumocytes, 1920 conidia, 15 macrophages, and 2160 iterations. PAI++ compiled with g++ 12.4.0 using the MinGW toolchain with level three optimization; jPAI run with Java 23.0.1. Execution environment: Windows 10 Education 64-Bit Dell laptop, Intel® Core™ Ultra 7 155U (14CPUs), ~2.1GHz, 16384MB RAM. A-B: paired t-test. C-D: Wilcoxon rank-sum test. (ns not significant $p > 0.05$; **** $p < 0.0001$).

# Benchmark

PAI++ contains only the code of the Ribeiro et al. (2022) model, which is the focus of this paper, while jPAI contains other pieces of code not discussed here. The code used for this benchmark is a simplification of Ribeiro et al. (2022). Within this context, PAI++ and jPAI are nearly identical (Figure 3A-B). The similarity between the two implementations serves as an important benchmark for model reproducibility, a key target in the modeling community (Donkin et al. (2017); Masison et al. (2021)). Figure 3C shows that the running time (wall time) of both implementations is similar (~11 sec), with C++ being slightly faster. On the other hand, there were major differences in the memory usage (Figure 3D). A more thorough benchmark is provided in Benchmark.pdf.

We provided documentation only for the Java version of the code. However, a C++ code follows the same structure.

# Output Description and Code Availability

The model outputs several values. For cells such as Aspergillus fumigatus and its several life stages (resting conidia, swelling conidia, germinating conidia, and hyphae), Macrophages, Neutrophils, and Type-II pneumocytes, the outputs are integer counts of these cells. For molecules (Lactoferin, Transferrin, and TAFC and its bound to iron forms, TNF-a, TGF-b, IL10, MIP1-b, MIP-2), the outputs are double floating point numbers representing the quantity of these molecules in mols (not concentration in molar). Concentration can be obtained by dividing these numbers by the simulated volume, 6.4e-8 liters. More details are provided in the documentation.

Code is available at: https://github.com/NutritionalLungImmunity/PAI.git

## AI Usage Disclosure

Generative AI tools were used to assist with paper writing and documentation. The software implementation, algorithmic design, benchmarking, and core technical contributions were developed without AI assistance. However, AI was used to assist with translating code from Java to C++, although there was extensive manual input in the form of coding, checking, and debugging. All AI-assisted content has been reviewed and validated by the authors for technical accuracy and scholarly integrity.

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